

CURRICULUM VITAE

Marco Díaz Maceda

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Short Bio

I am a theoretical physicist with two peer-reviewed publications. I conduct research as a PhD student at the Department of Theoretical Physics at Universidad Autónoma de Madrid (UAM). I seek to specialize in the application of geometrical techniques to quantum field theory.

I have contributed twice with scientific publications on my research in quantum computing and simulation of field theory systems in curved backgrounds during my BSc thesis.

I completed my MSc at the Instituto de Física Teórica (IFT) in Madrid. In my master's thesis, I formalized asymptotic symmetries in electrodynamics and General Relativity using the multisymplectic formalism.

I currently research about multisymplectic formalism for field theory in manifolds with boundaries and about the relationship between generalized symmetries and conserved charges of lower degree.

Publications

1. M. D. Maceda* and C. Sabín

Analogue simulation of quantum gravity black hole models in a dc-SQUID array

ARXIV: [arXiv:2507.14150](#)

PUBLISHED IN: Scientific Reports **15**, 20626 (2025)

DOI: [10.1038/s41598-025-07349-z](#)

PERSONAL CONTRIBUTION: I developed and executed the project, performed the computations and figures, and wrote the first version of the manuscript; **lead author**.

2. M. D. Maceda* and C. Sabín

Digital quantum simulation of cosmological particle creation with IBM quantum computers

ARXIV: [arXiv:2410.02412](#)

PUBLISHED IN: Scientific Reports **15**, 3476 (2025)

DOI: [10.1038/s41598-025-87015-6](#)

PERSONAL CONTRIBUTION: I made the computations, programmed and launched the experiments, analysed the data and wrote the first version of the manuscript; **lead author**.

FUNDING: My research was awarded by IBM with 12 hours of quantum computing, valued at 70,000 USD.

Education

MSc in Theoretical Physics

09.2024 – 07.2025

INSTITUTO DE FÍSICA TEÓRICA (IFT), Madrid, **Spain**

GPA: 9.09/10

SUPERVISOR: Javier Matulich

DATE OF DEFENCE: 02.07.2025

THESIS TITLE: An approach to asymptotic symmetries from multisymplectic formalism

BSc in Physics

09.2020 – 06.2024

UNIVERSIDAD AUTÓNOMA DE MADRID (UAM), Madrid, **Spain**

GPA: 8.71/10

SUPERVISOR: Carlos Sabín Lestayo

DATE OF DEFENCE: 02.06.2024

THESIS TITLE: Quantum simulation of curved spacetime (Cum Laude)

Research Experience

PhD research

06.2026 – Present

UNIVERSIDAD AUTÓNOMA DE MADRID (UAM), Madrid, **Spain**

TITLE: Covariant approach to observables in classical and quantum field theory

CO-SUPERVISOR: Carlos Sabín Lestayo

CO-SUPERVISOR: Jordi Gaset Rifà

FUNDER: Universidad Autónoma de Madrid (FPI-UAM)

PERSONAL CONTRIBUTION: I research about covariant quantization of field theory using multisymplectic formalism.

Member of working group

06.2025 – 05.2026

UNIVERSIDAD AUTÓNOMA DE MADRID (UAM), Madrid, **Spain**

TITLE: Quantum Gravitational Entanglement

REF NUMBER: CNS2024-154149

DATES: 04.2025–03.2027

PI: Carlos Sabín Lestayo

FUNDER: Ministerio de Ciencia, Innovación y Universidades

PERSONAL CONTRIBUTION: I conducted research on the efficient computability of quantum field theory using quantum computers and analogue systems.

Long research visit

06.2026 – 07.2026

UNIVERSITY OF WARSAW, Warsaw, **Poland**

HOST SUPERVISOR: Javier de Lucas Araujo

Research visit

05.2025 – Present

CUNEF UNIVERSIDAD, Madrid, **Spain**

HOST SUPERVISOR: Jordi Gaset Rifà

PERSONAL CONTRIBUTION: I conducted research on classical field theory in manifolds with boundaries within the multisymplectic formalism. I also developed a version of the Noether's theorem relating higher symmetries with conserved charges.

MSc research

06.2024 – 06.2025

INSTITUTO DE FÍSICA TEÓRICA (IFT), Madrid, **Spain**

SUPERVISOR: Javier Matulich

PERSONAL CONTRIBUTION: I investigated asymptotic symmetries within the multisymplectic formalism in electrodynamics and General Relativity.

BSc research and Collaboration Scholarship

09.2023 – 06.2024

UNIVERSIDAD AUTÓNOMA DE MADRID (UAM), Madrid, **Spain**

SUPERVISOR: Carlos Sabín

PERSONAL CONTRIBUTION: I developed simulations of quantum fields in curved spacetime using quantum computers and analogue systems.

Research fellow

06.2023 – 09.2023

INSTITUTO DE FÍSICA TEÓRICA (IFT), Madrid, **Spain**

SUPERVISOR: Juan F. Pedraza

PERSONAL CONTRIBUTION: I computed correlators in quantum field theories by calculating geodesics in higher-dimensional spaces within AdS/CFT.

Research fellow

09.2022 – 05.2023

INSTITUTO DE CIENCIA DE MATERIALES (ICMM), Madrid, **Spain**

SUPERVISOR: Gloria Platero

PERSONAL CONTRIBUTION: I designed unidimensional chains with useful topological properties.

Research Awards

PhD Scholarship (FPI-UAM)

06.2026 – 05.2030

FUNDER: Universidad Autónoma de Madrid (UAM)

BUDGET: 76,028 EUR

PROJECT TITLE: Covariant approach to observables in classical and quantum field theory

OBJECTIVE: Training of research personnel in the doctoral program of theoretical physics

Founding for registration

07.2026

CONFERENCE: 2nd International Workshop on Geometric Field Theory

BUDGET: Registration fee for attendance

Founding for registration

01.2026

CONFERENCE: VIII Iberoamerican Meeting on Geometry, Dynamics, and Field Theory

BUDGET: Registration fee for attendance

Summer School Scholarship

06.2025

SCHOOL: XVII International ICMAT Summer School on Geometry, Dynamics and Field Theory

BUDGET: Accommodation and fee for attendance

IBM Quantum Credits Grant

11.2023

FUNDER: International Business Machines (IBM).

BUDGET: 70,000 USD

PROJECT TITLE: Quantum simulation of curved spacetime

OBJECTIVE: Grant for researchers in quantum computing and simulation for quantum computers usage to advance the state-of-the-art of this discipline

Collaboration Scholarship

09.2023

FUNDER: Ministerio de Educación y Formación Profesional

BUDGET: 2,000 EUR

PROJECT TITLE: Quantum simulation of curved spacetime

OBJECTIVE: To support and enhance my research on quantum simulation

COMPETITIVE: Awarded to a single student from the entire department

Talks

Competition based short talk

07.2026

Bedlewo, **Poland**

TITLE: Multisymplectic formalism for field theory in manifolds with boundary

Seminar for the Department of Mathematical Methods in Physics

07.2026

UNIVERSITY OF WARSAW, Warsaw, **Poland**

TITLE: Multisymplectic formalism for field theory in manifolds with boundary

- Competition based short talk** **07.2026**
 UNIVERSITY OF WARSAW, Warsaw, **Poland**
 TITLE: Manifolds with boundaries, field theory and multisymplectic formalism
 AWARDS: Best short talk
- Competition based short talk** **06.2026**
 UNIVERSIDAD AUTÓNOMA DE MADRID, Miraflores de la Sierra, **Spain**
 TITLE: Field theory in manifolds with boundaries: a multisymplectic approach
- Competition based short talk** **03.2026**
 LOUGHBOROUGH UNIVERSITY, Loughborough, **United Kingdom**
 TITLE: Multisymplectic formalism for manifolds with boundaries
- Seminar for the Geometrical Mathematical Physics Research group** **03.2026**
 CUNEF, Madrid, **Spain**
 TITLE: Multisymplectic formalism for manifolds with boundaries
- Seminar for QMath** **03.2026**
 UNIVERSIDAD CARLOS III, Madrid, **Spain**
 TITLE: Multisymplectic formalism for manifolds with boundaries
- Seminar for the Geometrical Mathematical Physics Research group** **10.2025**
 CUNEF, Madrid, **Spain**
 TITLE: Kanatchikov's quantization: problems and perspective

Poster

- Poster exhibition at XVII International ICMAT Summer School on Geometry, Dynamics and Field Theory** **06.2025**
 INSTITUTO DE CIENCIAS MATEMÁTICAS (ICMAT), Madrid, **Spain**
 TITLE: A multisymplectic approach to boundary terms

Courses

- String theory doctorate course** **04.2026-05.2026**
 INSTITUTO DE FÍSICA TEÓRICA (IFT), Madrid, **Spain**
 LECTURER: Miguel Montero.
 COURSE OVERVIEW: Introduction to string theory, including bosonic string theory, the five supersymmetric string theories in ten spacetime dimensions, toroidal compactification, dualities, M-theory, D-branes and AdS/CFT holography.
- Reading groupoid seminar** **10.2025-06.2026**
 INSTITUTO DE CIENCIAS MATEMÁTICAS (ICMAT), Madrid, **Spain**
 LECTURERS: Oscar Carballal, Bartosz Maciej Zawora, Rubén Izquierdo López, Asier López Gordón, Juan Manuel López Medel.
 COURSE OVERVIEW: Systematic exploration of Lie groupoids, Lie algebroids, and their profound connections to modern differential geometry and mathematical physics.

Outreach

European Researchers' Night

09.2026

CÍRCULO DE BELLAS ARTES, Madrid, [Spain](#)

Fifth Vicálvaro Scientific Monologue Competition

04.2018

UNIVERSIDAD REY JUAN CARLOS, Madrid, [Spain](#)

CONTRIBUTION: I presented a monologue about special relativity.

Fourth Vicálvaro Scientific Monologue Competition

04.2017

UNIVERSIDAD REY JUAN CARLOS, Madrid, [Spain](#)

CONTRIBUTION: I presented a monologue about the Standard Model of particle physics.

PRIZE: Award for the monologue with the best scientific content.